Internet Access and Household Income in South Carolina

Econometrics Final Project
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Motivation: Recent Investment In SC

Private investment for Broadband in Rural Areas

- Connect America Fund (2015-1019) (Ongoing)
- FCC Rural Digital Opportunity Fund (2020) (100M)
- 400M in Federal Funds and 50M in state funds

Public Investment for Broadband in Rural Areas

- AT&T $825M (2016-2018)
- Comporium awarded $51.9 Million (2023)

Research Question

What is the relationship between broadband internet access and household income in rural and urban south carolina counties?

Methodology and Data

Panel Data Analysis

- Results of development infrastructure initiatives take time to reveal themselves , thus a panel data analysis would hopefully capture some of the results(SOURCE)

Example

Charleston has been an affluent county for a long time, and to compare financial growth without taking into account the different county specific financial history and industries would be unfair comparison.

ACS-5 vs. ACS-1 (US Census B.)

ACS-5:

- Average specific data from a 5 year period. Includes all counties.

ACS-1:

- Annual Data Collected on counties with big populations

Justification

ACS-5 files were chosen because they provided data for ALL counties in south carolina. Thus, making my research less biased towards counties with bigger populations with more infrastructure

Variables Used

Broadband_pct:
Percentage of all households in the county with broadband, on average in the last 5 years

Median_inc:
- The median household income for each county

Pop_total:
- Total population in county

Year:
- Factored for invariant time effects

County:
- Factored for differences in county wealth and industry

County Type:
- Urban : Counties with populations over 100,000, Rural otherwise

- I logged all variables except for year, county, and county_type

State Wide Fixed Effects Model

$$\log(\text{median\_inc}) = \beta_0 + \beta_1 \cdot \log(\text{broadband\_pct}) + \beta_2 \cdot \log(\text{pop\_total}) + \text{County FE} + \text{Year FE} + \varepsilon$$

term <chr>	estimate <dbl>	p.value <dbl>
(Intercept)	7.3828253	2.689696e-04
log(broadband_pct)	0.1279428	1.350181e-01
log(pop_total)	0.2593903	1.379512e-01
factor(year)2022	0.2370578	4.633266e-17

Rural Fixed Effects Model

$$\log(median\_inc) = \beta_0 + \beta_1 \cdot \log(broadband\_pct) + \beta_2 \cdot \log(pop\_total) + \text{County FE} + \text{Year FE} + \varepsilon$$

term	estimate	p.value
<chr>	<dbl>	<dbl>
(Intercept)	6.9391848	2.813876e-02
log(broadband_pct)	0.1376083	1.923729e-01
log(pop_total)	0.2993649	3.047084e-01
factor(year)2022	0.2376760	4.868966e-10

Urban Fixed Effects Model

$$\log(\text{median\_inc}) = \beta_0 + \beta_1 \cdot \log(\text{broadband\_pct}) + \beta_2 \cdot \log(\text{pop\_total}) + \text{County FE} + \text{Year FE} + \varepsilon$$

1. Negative log(Broadband_pct) coefficient?

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
(Intercept)	7.35169837	3.33121479	2.2069121	4.754331e-02
log(broadband_pct)	-0.06836938	0.22227998	-0.3075823	7.636731e-01
log(pop_total)	0.30926331	0.28114140	1.1000276	2.928982e-01
factor(year)2022	0.25070385	0.02825232	8.8737437	1.281414e-06

2. Why?

Urban Fixed Effects Model

$$\log(median\_inc) = \beta_0 + \beta_1 \cdot \log(broadband\_pct) + \beta_2 \cdot \log(pop\_total) + \text{County FE} + \varepsilon$$

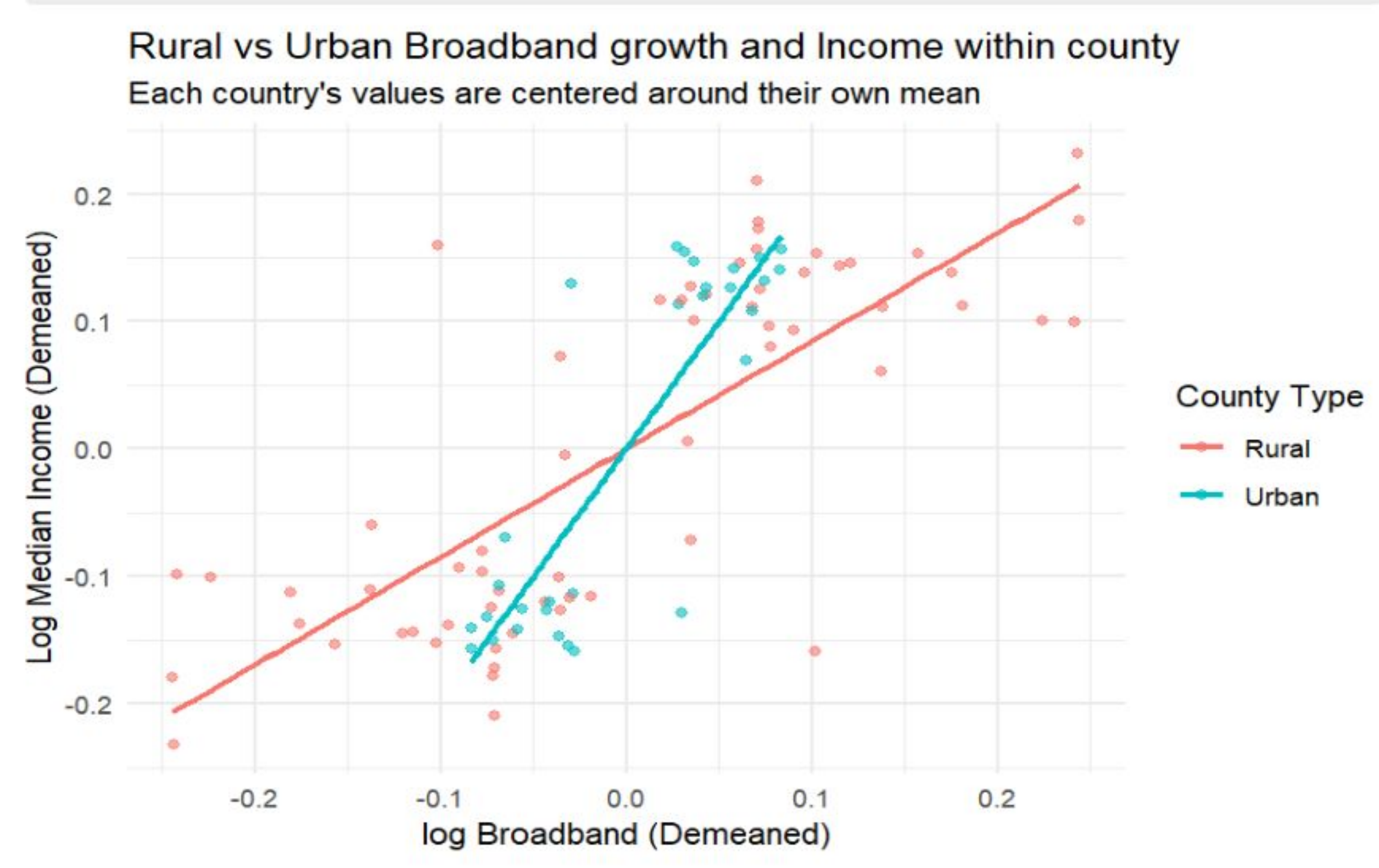
1. Removed the factor(year)

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
(Intercept)	12.2591561	6.5854378	-1.861555	0.08542397
log(broadband_pct)	1.2404757	0.4393478	2.823448	0.01436982
log(pop_total)	1.4978751	0.6530577	2.293634	0.03911807

2. Why?

Potentially... Hierarchical Diffusion (Rogers (2003))

If all urban counties increase in broadband connection simultaneously , factoring for year would remove the impact of of the simultaneous increase


Rural vs Urban Broadband growth and Income within county
Each country's values are centered around their own mean
Log Median Income (Demeaned)
0.2
0.1
0.0
-0.1
-0.2
-0.2
-0.1
0.0
0.1
0.2
log Broadband (Demeaned)
County Type
Rural
Urban

Answer

Urban

Median Household income and Internet access are positively correlated with each other

Rural

Median Household income and internet access are positively correlated with each other

Significant?

Causal?

Research Question:

What is the relationship between internet access and income in rural and Urban Counties in South Carolina?

Omitted Variables

Educational Attainment

Unemployment Rate

Age

Next Steps

- Account for educational Attainment in models
- Account for unemployment rates in model
- Account for age in models

Sources

U.S. Census Bureau. (n.d.). *Comparative demographic estimates*. American Community Survey, ACS 5-year estimates comparison profiles, Table CP05. Retrieved April 14, 2026, from https://data.census.gov/table/ACSCP5Y2017.CP05?t=Older+Population:Population+Total:Populations+and+People&g=040XX00US45$0500000

U.S. Census Bureau, U.S. Department of Commerce. (n.d.). *Educational attainment*. American Community Survey, ACS 5-year estimates subject tables, Table S1501. Retrieved April 14, 2026, from https://data.census.gov/table/ACSST5Y2023.S1501?q=Education&g=040XX00US45$0500000

U.S. Census Bureau. (n.d.). *Types of computers and internet subscriptions*. American Community Survey, ACS 5-year estimates subject tables, Table S2801. Retrieved April 14, 2026, from https://data.census.gov/table/ACSST5Y2022.S2801?q=broadband&t=Educational+Attainment&g=040XX00US45$0500000

U.S. Census Bureau. (n.d.). *Income in the past 12 months (in 2017 inflation-adjusted dollars)*. American Community Survey, ACS 5-year estimates subject tables, Table S1901. Retrieved April 14, 2026, from https://data.census.gov/table/ACSST5Y2017.S1901?t=Income+(Households,+Families,+Individuals)&g=040XX00US45$0500000

Rogers, E. M. (2003). *Diffusion of innovations* (5th ed.). Free Press.

Universal Service Administrative Company. (n.d.). *Connect America Fund Phase II*. Retrieved April 15, 2026, from https://www.usac.org/high-cost/funds/caf-phase-ii/

Sources

Universal Service Administrative Company. (n.d.). *Rural Digital Opportunity Fund*. Retrieved April 15, 2026, from https://www.usac.org/high-cost/funds/rural-digital-opportunity-fund/

South Carolina Office of Regulatory Staff. (n.d.). *SC Broadband Office completes historic $400 million American Rescue Plan commitment*. Retrieved April 15, 2026, from https://ors.sc.gov/news/scbbo-completes-historic-400-million-american-rescue-plan-commitment-connect-over-112380

Comporium. (n.d.). *Comporium awarded $51.9 million in grants by the SC Broadband Office*. Retrieved April 15, 2026, from https://www.comporium.com/about_news/comporium-awarded-51-9-million-in-grants-by-the-sc-broadband-office/

AT&T. (2019, June 24). *AT&T invests nearly $825 million over 3-year period to boost local networks in South Carolina*. PR Newswire. https://www.prnewswire.com/news-releases/att-invests-nearly-825-million-over-3-year-period-to-boost-local-networks-in-south-carolina-300872407.html